day-01-interview-qa my answer

Tags	
Topic	python notes

1. List vs tuple vs set vs dict: when do you use each?

List is used when I need mutable, duplicated,indexed values to store. set is used when I need unique values to store where index doesn't matter. tuple is need when I need immutable,indexed,duplicated values to store. dict is needed when I need to store values as key value pairs.

1. Why is `x in my_set` faster than `x in my_list`?

set has non duplicated values thats why its faster that list values.

1. `sort()` vs `sorted()`.

sort() and sorted() both used to sort iterable objects. The main difference is sort changes the original array or list but sorted() creates a new object and doesn't mutate the original one.

1. `dict["key"]` vs `dict.get("key")`.

If a dictionary value is sure that thats there, dict[”key”] can be used, It will give error if that value is not there. But dict.get(”key”) won't give error if there is no value, it'll give none.

1. List comprehension vs generator expression: what's the memory difference?

List comprehension remebers all the items in the list thats why the memory gets bigger, but generator use it only when needed and also generator remembers the previous state. Thats why memory takes low space.

1. What's the difference between an iterable, an iterator, and a generator?

Simply speaking, when an object can loop its called iterable, and iterator is that state or bookmark that object left when iterating. Generator is another king of iterator, but it remembers its previous state and memory efficient.

1. What is variable shadowing, and why is naming a variable `sum` a problem?" Use your own `sum = 0` as the example.
2. "How would you count items or group items by a key in Python?" Answer with your `Counter` and `defaultdict` code.
3. What's the difference between a shallow copy and a deep copy?

shallow copy doesn't work copying a mutable object like dictionary, deep copy works.

1. How do you flatten a list of lists in Python?

1. When would you NOT use a list comprehension?

when reading a big 6gb file or reading csv files that are too big.

1. What does `a, b = b, a + b` do, and why doesn't `a = b; b = a + b` give the same result?